

Summary

AI Robotics Engineer with an M.S. in Electrical Engineering, focused on ROS 2, Isaac Sim / Isaac Lab, and Isaac GROOT for robot control, VLA, and Digital Twin data workflows. Experienced in end-to-end integration across perception, visual servoing, contact-aware insertion, data processing, policy inference, frontend/backend platforms, and simulation / real-world validation.

Key Results

3-4x imitation-learning / GROOT-VLA training-data throughput	70% → 90% cable-insertion success in 10 trials
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Experience

AMTRAN TECHNOLOGY CO., LTD. | AI Robotics Engineer May 2025 - Present | New Taipei, Taiwan

- Built ROS 2 control and simulation workflows for industrial and collaborative robot platforms with NVIDIA Isaac Sim / Isaac Lab, MoveIt 2, and cuMotion for motion planning, collision handling, and cable tasks.
- Delivered Isaac GROOT N1.7 data conversion, model training, and policy inference, separately validating virtual-data training followed by real-world inference and real-data training/inference workflows.
- Built a two-part training-data collection flow that automated robot tasks and recording through ROS 2, WebSocket, and related interfaces, fully automating multiple imitation-learning / GROOT / VLA training episodes and achieving 3-4x manual throughput.
- Owned robotics software from requirements and architecture through Docker deployment, testing, on-site integration, and maintenance; integrated YOLO OBB / OpenCV visual servoing for HDMI insertion and AprilTag-based connector localization for cable tasks, improving cable-insertion success from 70% to 90% in a 10-trial test.
- Integrated contact-aware insertion alignment with visual servoing and force-guided search, and validated the complete workflow on the physical robot.

ISCOM | Software Engineer Intern Feb 2022 - Jun 2022 | Kaohsiung, Taiwan

- Participated in backend API development, frontend integration, and web feature implementation using C#, ASP.NET MVC, JavaScript, jQuery, MS SQL, and Git.

Skills

Robot Control & Digital Twin	ROS 2, NVIDIA Isaac Sim / Lab, MoveIt 2, cuMotion, virtual / physical robot workflows
VLA & Robot Learning	Isaac GROOT N1.7, VLA, Policy Inference, Imitation Learning, Robot State / Action
Perception & Visual Inspection	RealSense, YOLO OBB + OpenCV, AprilTag + OpenCV, Hand-Eye Calibration, Visual Servoing, contact-aware insertion, PatchCore anomaly detection with PyQt
Software & System Integration	Python, C++, frontend/backend platforms with FastAPI, React / TypeScript, Next.js, WebSocket, Docker, Linux, Arduino, ESP32, MQTT

Education & Honors

- M.S. in Electrical Engineering, National Taiwan Normal University | Sep 2022 - Jul 2024
- B.S. in Computer and Communication, National Pingtung University | Sep 2018 - Jun 2022
- Honors: RHCSA; FIRA Roboworld Cup First Place; National Technical Creativity Competition Third Place